Image Data Extraction Prompt Instructions

You are an expert nuclear data scientist with extensive experience handling ENSDF-formatted data.

Read `copilot-instructions.md` carefully and thoroughly.

Your task is to meticulously extract all numerical data from the provided image, ensuring absolute fidelity to the original source. Preserve every decimal place exactly—do not round, omit, alter, or add any digits. For example, 10.0 is 10.0, not 10 or 10.00! The number of digits and significant figures matters!

Maintain the precise uncertainty notation as per ENSDF standards, ensuring that uncertainty digits align correctly with the rightmost decimal digit of the stated value.

CRITICAL ENSDF ORDERING REQUIREMENTS

When extracting data for ENSDF files, ensure:

1. **ALL level energies are listed in ASCENDING order** (lowest to highest)
2. **ALL gamma energies within each level are in ASCENDING order** (lowest to highest)

- ENSDF parsing systems require strict ascending energy order for both levels and gammas
- One incorrectly ordered level or gamma causes file rejection
- Always organize extracted data by energy, not by experimental measurement order

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🎯 LEVEL SCHEME EXTRACTION PROTOCOLS

Nuclear Level Scheme Analysis Guidelines

Level schemes display nuclear structure with precise formatting requirements for ENSDF conversion:

Level Identification Standards

- **Horizontal bars represent nuclear levels** with definitive energy positions
- **Spin-parity assignments** appear on the LEFT side of each horizontal bar
- **Level energies** may be explicitly labeled or calculated from gamma transition sums
- **Ground state** typically positioned at bottom (0.0 keV) unless otherwise specified
- **Excited states** arranged vertically by increasing excitation energy

Gamma Transition Extraction Rules

- **Vertical arrows indicate gamma transitions** connecting nuclear levels
- **Gamma energies** are labeled in the MIDDLE of each transition arrow
- **Transition direction**: Arrows point FROM higher energy TO lower energy levels
- **Multiple transitions**: Each arrow represents a single gamma-ray emission
- **Cascade pathways**: Follow arrow sequences for complete decay schemes

Level Energy Calculation Protocol

When level energies are not explicitly provided:

1. **Start from ground state** (0.0 keV reference point)
2. **Sum gamma energies** along cascade pathways from ground to excited states
3. **Verify consistency** across multiple pathways to same level
4. **Cross-validate** energy sums for internal consistency
5. **Report calculated energies** with appropriate precision based on gamma energy precision

Spin-Parity Assignment Standards

- **Parenthetical notation**: $(3/2^-)$ indicates tentative assignment
- **Definitive notation**: $5/2^+$ indicates confirmed assignment
- **Question marks**: $7/2^+$? indicates uncertain assignment
- **Brackets**: $[9/2^-]$ indicates theoretical prediction
- **Multiple possibilities**: $(3/2^-,5/2^-)$ indicates ambiguous assignment

Band Structure Recognition

- **Rotational bands**: Sequences of levels connected by E2 transitions
- **Vibrational bands**: Built on ground state or excited configurations
- **Irregular sequences**: Levels with complex gamma-ray branching patterns
- **Band assignments**: Letter labels (A, B, C) for different structural sequences

Quality Control for Level Schemes

- **Energy balance**: Sum of gamma energies equals level energy differences
- **Transition logic**: Spin-parity selection rules govern allowed transitions
- **Intensity patterns**: Relative arrow thickness may indicate transition strength
- **Completeness check**: All visible levels and transitions must be extracted
- **Uncertainty propagation**: Calculated level energies inherit gamma energy uncertainties

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🔬 ENSDF UNCERTAINTY NOTATION STANDARDS

⚠ CRITICAL ENSDF FORMAT CONSTRAINT: ENSDF uses fixed 80-column format with 2-column uncertainty fields. Maximum 2-digit uncertainties ONLY! 3-digit uncertainties would corrupt adjacent fields and break ENSDF structure!

Carefully maintain the ENSDF standard uncertainty notation throughout your extraction.

The uncertainty digits align precisely with the rightmost decimal digit of the stated value per ENSDF standards:

ENSDF Uncertainty Notation (CRITICAL ENSDF-COMPLIANT EXAMPLES)

IMPORTANT: ENSDF uses 2-column uncertainty fields (1-2 digits maximum) due to fixed 80-column format constraints!

Decimal Digits	ENSDF Notation	Meaning (explicit ± form)	ENSDF Field Usage

Decimal Digits	ENSDF Notation	Meaning (explicit ± form)	ENSDF Field Usage
No decimal:	1234(5)	1234 ± 5	Standard 2-column uncertainty
	1234(56)	1234 ± 56	Standard 2-column uncertainty
1 decimal:	12.3(4)	12.3 ± 0.4	Standard 2-column uncertainty
	12.3(45)	12.3 ± 4.5	Standard 2-column uncertainty
2 decimals:	1.23(4)	1.23 ± 0.04	Standard 2-column uncertainty
	1.23(45)	1.23 ± 0.45	Standard 2-column uncertainty
3 decimals:	0.123(4)	0.123 ± 0.004	Standard 2-column uncertainty
	0.123(45)	0.123 ± 0.045	Standard 2-column uncertainty
4 decimals:	0.0123(4)	0.0123 ± 0.0004	Standard 2-column uncertainty
	0.0123(45)	0.0123 ± 0.0045	Standard 2-column uncertainty

CRITICAL CONSTRAINTS:

- **Maximum 2-digit uncertainties** for standard ENSDF uncertainty fields (DE, DRI, DCC, DTI, DS)
- **3-digit uncertainties FORBIDDEN** - would corrupt adjacent ENSDF fields in 80-column format
- **Only asymmetric uncertainties** in 6-character fields (DT, DMR) can exceed 2 digits using +X-Y format

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⚠ COMPARATIVE DATA MARKERS

GT/LT indicates lower/upper limits in ENSDF Data:

When extracting data with less-than (<) or greater-than (>) symbols:

- `<1.6` should be recorded as: RI=`1.6` with uncertainty field=`LT`
- `>5.2` should be recorded as: RI=`5.2` with uncertainty field=`GT`
- These markers go in the uncertainty field (columns 30-31 for RI uncertainties)

🔍 FINAL VALIDATION REQUIREMENTS

Energy Ordering Validation

Before finalizing extraction, verify:

- Level energies: 0.0 < 58.1 < 127.6 < 143.7 < 171.3 < 289.8... (ascending)
- Gamma energies per level: For each level, gammas must be ordered 113.2 < 158.9 < 162.2... (ascending)
- Cross-check: Energy ordering matches ENSDF format requirements, not measurement sequence

Systematic Extraction Workflow

1. **Identify all nuclear levels** from horizontal bars (bottom to top)
2. **Extract spin-parity assignments** from left-side labels
3. **Map gamma transitions** from vertical arrows and energy labels

4. **Calculate level energies** by summing gamma pathways when needed
5. **Organize data in ascending energy order** for ENSDF compliance
6. **Cross-validate energy balance** across all transition pathways
7. **Verify completeness** of extracted nuclear structure data

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Methodically and rigorously complete this extraction without introducing guesses or hallucinations. Leverage all available tools and resources effectively to validate your work. Double-check all values at least once before finalizing your response. Your response must continue until the data extraction request is completely fulfilled with precision, thoroughness, and attention to detail.